Python Practice Set: Conditionals

Covers: variables, data types, print(), input(), string methods, and if / elif / else / match-case.

Part 1 — Movie Ticket Counter (if / elif / else)

You're building a simple ticket counter for a small cinema. Write a program that does the following:

- Ask the user to enter their **name** using input().
- Ask the user to enter their **age** using input(), and convert it to an integer.
- Ask the user to enter the **day** they're booking for (e.g. "Monday", "Saturday").
- Decide the ticket price using if / elif / else based on age:
- Age under 5 → price is 0 (free).
- Age 5 to 12 (inclusive) → price is 100.
- Age 13 to 59 (inclusive) → price is 200.
- Age 60 or above → price is 120 (senior discount).

Next, check the day the user entered (use a string method so it doesn't matter if they typed "saturday", "SATURDAY" or "Saturday"). If the day is Saturday or Sunday, add 50 to the price as a weekend surcharge.

Finally, print a clean summary message using the person's name (also clean up extra spaces from the name using a string method), something like:

> Hello Aman, your ticket for Saturday costs 250 rupees.

Sample run to match:

```
Enter your name: aman
Enter your age: 17
Enter the day: saturday
Hello Aman, your ticket for Saturday costs 250 rupees.
```

> Hint: str.strip() removes extra spaces, str.title() or str.capitalize() fixes the casing of the name, and str.lower() is useful for comparing the day without worrying about case.

Part 2 — Simple Login Gate (if / elif / else)

Extend the same program (or write a new one) that checks login details before showing the ticket summary from Part 1:

- Store a fixed correct username and password as variables, for example: correct_user = "admin" and correct_pass = "cinema123".
- Ask the user to input a username and a password.
- If both match exactly, print "Access granted" and continue to the ticket logic from Part 1.

- If the username is correct but the password is wrong, print "Wrong password".
- If the username itself is wrong, print "User not found".

  Use elif so only one of the three messages ever prints, never more than one.

Part 3 — Grocery Bill Category (match-case)

Now practice match-case, Python's pattern-matching version of if/elif. Build a small grocery discount checker:

- Ask the user to input an item name, e.g. "rice", "milk", "soap", "chocolate".
- Use a string method to make the input lowercase before matching, so "Rice" and "rice" behave the same way.
- Use match-case on the item name to decide its category and discount:
- "rice" or "wheat" or "dal" → category "Staples", discount 10%
- "milk" or "curd" or "paneer" → category "Dairy", discount 5%
- "soap" or "shampoo" → category "Personal Care", discount 8%
- anything else → category "General", discount 0% (use the case _ : pattern for this)

  Tip: in match-case you can group several values in one case using the pipe symbol, like case "milk" | "curd" | "paneer":

Then ask for the item's price (as a float using input()), calculate the discount amount and the final price, and print a receipt-style summary, for example:

```
Enter item name: Milk
Enter price: 60
Item: Milk
Category: Dairy
Discount: 5%
You pay: 57.0
```

Part 4 — Put It Together: Mini Billing System

Combine everything above into one program with this flow:

- 1. Login check from Part 2 (if / elif / else). Only continue if access is granted.
- 2. Once logged in, ask how many items the customer wants to bill, e.g. n = int(input(...)).
- 3. Repeat the Part 3 match-case logic for each item (you can copy-paste the match block inside a loop if you already know loops, otherwise it's fine to just do it once per run for now).
- 4. Keep a running total variable that adds up each item's final price.
- 5. At the end, print the total bill. If the total is above 500, print an extra line: "You qualify for free home delivery!" using one more if condition.

Checklist before you consider it done

- Program runs without errors for normal input.
- Age boundaries (exactly 5, exactly 12, exactly 13, exactly 59, exactly 60) all give the correct price.
- Day comparison works regardless of how the user types the day (case-insensitive).
- Login correctly distinguishes between wrong username vs wrong password.
- match-case correctly falls back to "General" for an item not listed.
- Final total and the delivery message print correctly.